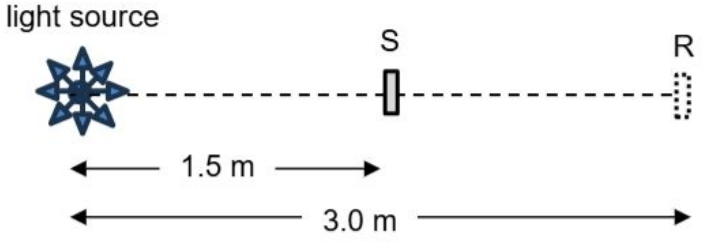
15 A light source radiates light in all directions.

A small surface is placed at position S a distance 1.5 m from the light source. The plane of the surface is perpendicular to the direction of incidence of the light from the light source. The surface receives 1.4 W of power from the light of the source.



The power of the light source is now doubled and the surface is moved to position R a distance 3.0 m from the light source. The plane of the surface is still maintained perpendicularly to the direction of incidence of the light from the light source.

What is the power received by the surface at position R?

	A	0.35 W	**B**	0.70 W	**C**	1.4 W	**D**	2.8 W

L2 **Answer: B**

Let P_s be the power of the source initially.
Intensity at distance r from the source $I = P_s/4\pi r^2$

Power received by surface S,
$P = I \times A$ **where A = area of the surface**
$= (P_s/4\pi r^2) \times A$
$P \propto (P_s / r^2)$ **A is constant**

Power of the source is doubled and r is also doubled. Hence,
(Power received at R) / (Power received at S) = $2 / 2^2 = ½$
Power received by R = ½ x 1.4 = 0.70 W